

WEEK 15

**WRITING BASIC
SQL SELECT STATEMENTS**

Objectives

After completing this lecture, you should be able to do the following:

Understanding of Basic SQL Statements.

Using different arithmetic operators inside SQL statements for data retrieval.

SQL

SQL stands for **Structured Query Language**

SQL lets you access and manipulate databases

BASIC SQL COMMANDS

There are a number of SQL statements, few of which are explained below:

- **Data Retrieval Statement:** SELECT is the data extracting statement which retrieves the data from the database.
- **Data Manipulation Language (DML):** This language constitutes the statements that are used to manipulate with the data. It has three commands, which are INSERT, UPDATE and DELETE.
- **Data Definition Language (DDL):** This is the language used to define the tables. It sets up, changes, and removes data structures from the tables. It uses 5 commands, which are CREATE, ALTER, DROP, RENAME and TRUNCATE.

Writing SQL Statements

SQL statements are not case sensitive.

SQL statements can be on one or more lines.

Keywords cannot be abbreviated or split across lines.

Clauses are usually placed on separate lines.

Indents are used to enhance readability.

Basic SELECT Statement

```
SELECT    *|{[DISTINCT] column|expression [alias],...}  
FROM      table;
```

SELECT identifies *what* columns

FROM identifies *which* table

In the syntax:

KEYWORD	MEANING
SELECT	is a list of one or more columns
*	selects all columns
DISTINCT	suppresses duplicates
column expression	selects the named column or the expression
alias	gives selected columns different headings
FROM table	specifies the table containing the columns

Selecting All Columns

```
SELECT *  
FROM departments;
```

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500
90	Executive	100	1700
110	Accounting	205	1700
190	Contracting		1700

8 rows selected.

Selecting Specific Columns

```
SELECT department_id, location_id  
FROM departments;
```

DEPARTMENT_ID	LOCATION_ID
10	1700
20	1800
50	1500
60	1400
80	2500
90	1700
110	1700
190	1700

8 rows selected.

Arithmetic Expressions

Create expressions with number and date data by using arithmetic operators.

Operator	Description
+	Add
-	Subtract
*	Multiply
/	Divide

Using Arithmetic Operators

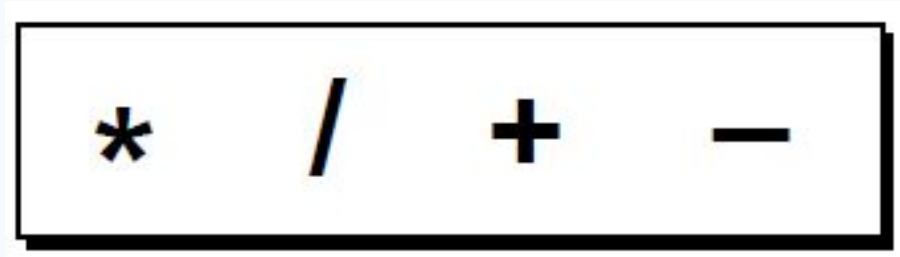
```
SELECT last_name, salary, salary + 300  
FROM   employees;
```

LAST_NAME	SALARY	SALARY+300
King	24000	24300
Kochhar	17000	17300
De Haan	17000	17300
Hunold	9000	9300
Ernst	6000	6300
Lorentz	4200	4500

Note:

Resultant calculated column SALARY+300 is not a new column in the EMPLOYEES table; it is for display only. By default, the name of a new column comes from the calculation that generated it— in this case, salary+300.

Operator Precedence



Multiplication and division take priority over addition and subtraction

Operators of the same priority are evaluated from left to right

Parentheses are used to force prioritized evaluation and to clarify statements

Operator Precedence

```
SELECT last_name, salary, 12*salary+100  
FROM   employees;
```

LAST_NAME	SALARY	12*SALARY+100
King	24000	288100
Kochhar	17000	204100
De Haan	17000	204100
Hunold	9000	108100
Ernst	6000	72100
Lorentz	4200	50500
Gietz	8300	99700

20 rows selected.

Using Parentheses

```
SELECT last_name, salary, 12*(salary+100)
FROM   employees;
```

LAST_NAME	SALARY	12*(SALARY+100)
King	24000	289200
Kochhar	17000	205200
De Haan	17000	205200
Hunold	9000	109200
Ernst	6000	73200
Lorentz	4200	51600
Gietz	8300	100800

20 rows selected.

Defining a Null Value

A null is a value that is unavailable, unassigned, unknown, or inapplicable

A null is not the same as zero or a blank space

```
SELECT last_name, job_id, salary, commission_pct
FROM   employees;
```

LAST_NAME	JOB_ID	SALARY	COMMISSION_PCT
King	AD_PRES	24000	
Kochhar	AD_VP	17000	
Zlotkey	SA_MAN	10500	.2
Abel	SA_REP	11000	.3
Taylor	SA_REP	8500	.2
Maggins	AC_MGR	9500	
Gietz	AC_ACCOUNT	8300	

20 rows selected.

Null Values in Arithmetic Expressions

Arithmetic expressions containing a null value evaluate to null.

```
SELECT last_name, 12*salary*commission_pct
FROM   employees;
```

LAST_NAME	12*SALARY*COMMISSION_PCT
King	
Kochhar	
Zlotkey	25200
Abel	39600
Taylor	20640
Higgins	
Gietz	

20 rows selected.

Null VALUES (Cont..)

If any column value in an arithmetic expression is null, the result is null.

For example, if you attempt to perform division with zero, you get an error.

However, if you divide a number by null, the result is a null or unknown.

Defining a Column Alias

A column alias:

- Renames a column heading

- Is useful with calculations

- Immediately follows the column name: there can also be the optional AS keyword between the column name and alias

- Requires double quotation marks if it contains spaces or special characters or is case sensitive

Using Column Aliases

```
SELECT last_name AS name, commission_pct comm
FROM   employees;
```

NAME	COMM
King	
Kochhar	
Higgins	
Gietz	

20 rows selected.

```
SELECT last_name "Name",
       salary*12 "Annual Salary"
FROM   employees;
```

Name	Annual Salary
King	288000
Kochhar	204000
Higgins	144000
Gietz	93600

20 rows selected.

Concatenation Operator

A concatenation operator:

Concatenates columns or character strings to other columns

Is represented by two vertical bars (||)

Creates a resultant column that is a character expression

Using the Concatenation Operator

```
SELECT last_name||job_id AS "Employees"  
FROM employees;
```

Employees
KingAD_PRES
KochharAD_VP
De HaanAD_VP
HunoldIT_PROG
GietzAC_ACCOUNT

20 rows selected

Literal Character Strings

A literal value is a character, a number, or a date included in the SELECT list

Date and character literal values must be enclosed within single quotation marks

Each character string is output once for each row returned

Using Literal Character Strings

```
SELECT last_name || ' is a ' || job_id  
       AS "Employee Details"  
FROM   employees;
```

Employee Details
King is a AD_PRES
Kochhar is a AD_VP
De Haan is a AD_VP
Hunold is a IT_PROG
Ernst is a IT_PROG
Gietz is a AC_ACCOUNT

20 rows selected.

Using Literal Character Strings

```
SELECT last_name ||': 1 Month salary = '||salary Monthly  
FROM   employees;
```

MONTHLY
King: 1 Month salary = 24000
Kochhar: 1 Month salary = 17000
De Haan: 1 Month salary = 17000
Hunold: 1 Month salary = 9000
Ernst: 1 Month salary = 6000
Lorentz: 1 Month salary = 4200
Mourgos: 1 Month salary = 5800
Rajs: 1 Month salary = 3500

Duplicate Rows

The default display of queries is all rows, including duplicate rows.

```
SELECT department_id  
FROM   employees;
```

DEPARTMENT_ID	
	90
	90
	90
	60
	60
	60
	50
	50

20 rows selected.

Eliminating Duplicate Rows

Eliminate duplicate rows by using the **DISTINCT** keyword in the **SELECT** clause.

```
SELECT DISTINCT department_id  
FROM   employees;
```

DEPARTMENT_ID	
	10
	20
	50
	60
	80
	90
	110

8 rows selected.

Displaying Table Structure

```
DESCRIBE employees
```

Name	Null?	Type
EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

View OBJECTS AND tables in database

To view the objects in the database

```
SELECT * FROM CAT;
```

To view the tables in the database

```
SELECT * FROM TAB;
```

Q/A